

Scalability & Future Plan

Date	18 July 2026
Team ID	TBD
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports limited concurrent users	Performance may decrease under heavy traffic	High
2	Uses SQLite database	Not suitable for very large datasets	Medium
3	Requires internet connection for Gemini API	AI feedback unavailable without internet	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-server deployment	Deploy on cloud with load balancing
2	Data Storage	SQLite Database	Upgrade to PostgreSQL or MySQL
3	Performance	Basic processing	Optimize models and enable asynchronous processing
4	Security	Basic API key authentication	Add user authentication, HTTPS, and role-based access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	User Login & Authentication	1 Month	Secure access and personalized reports
Phase 2	Multi-language Speech Evaluation	2 Months	Support regional languages
Phase 3	Cloud Deployment & Mobile App	3 Months	Better accessibility and scalability
Phase 4	Advanced AI Analytics Dashboard	4 Months	Detailed insights and improved learning experience